

Practix

A voice-native, offline-first clinical workspace for dentistry — product, compliance, operations and commercial record

PLATFORM	ARCHITECTURE	JURISDICTIONS	OWNER & DEVELOPER
Android API 24–36 & web portal	Offline-first, local sovereignty	HIPAA · GDPR · DPDP 2023	Devanapally Charan Tej

Dental clinicians spend a quarter to a third of chairside time typing. Practix removes the keyboard from the operatory: speech is transcribed on the device, clinical reasoning runs on the device, and the record never leaves the device unless a practice chooses otherwise. This dossier consolidates the four pre-release suites — product and architecture, legal and compliance, operations and quality, business and intellectual property — into one reviewable record.

Three frictions define the problem. Switching between instruments and a keyboard breaks sterile protocol. Charting deferred to the end of clinic hours produces recall error, documentation backlog and fatigue. And cloud-only practice systems place protected health information behind a network that may be slow, absent, or non-compliant with the mandate of the jurisdiction the chair sits in.

The answer assembled here is a cryptographic workspace: *whisper.cpp* for chairside dictation, *llama.cpp* GGUF runtimes for decision support, Google Filament for a three-dimensional odontogram, and SQLCipher keyed from a hardware-backed keystore — with an append-only, hash-chained audit trail underneath all of it.

CONTENTS

I	Product & Architecture	Requirements, personas, layered architecture, data model, sync contract
II	Legal & Compliance	Privacy posture, licence terms, clinical disclaimers, store declarations
III	Operations, QA & Delivery	Sprint plan, verification matrix, key lifecycle, accessibility
IV	Business & Intellectual Property	Segments, tiers, unit economics, assignment and confidentiality

1 · THE THREE FRICTIONS

Infection control

Moving between instruments and a shared keyboard or mouse breaks the sterile chairside field.

Note latency

Charting postponed to the end of clinic produces recall error, backlog and clinician burnout.

Connectivity fragility

Cloud-only systems risk latency spikes, outages and non-compliance while handling PHI.

2 · PERSONAS

CLINICIAN	ROLE & ENVIRONMENT	CORE PAIN POINT AND OBJECTIVE
Dr. Ingrid Halvorsen	Principal dental surgeon and practice administrator, private multi-chair clinic	Immutable audit logging, practice oversight, team invitation control, strict compliance evidence.
Dr. Tomas Ferreira	Associate dentist and periodontist, high-throughput operatory	Fast hands-free charting while gloved; instant drug interaction checks before issuing.
Sarah Jenkins, RDH	Registered dental hygienist, hygiene and prophylactic operatory	Rapid pocket-depth logging, bleeding index recording, automated patient summary export.

3 · FUNCTIONAL REQUIREMENTS

FR-1

Voice charting engine

Continuous audio streaming from `AudioRecordManager` into native `whisper.cpp` inference; a deterministic regex and semantic parser covering dental vernacular (“Tooth 16 MO composite restoration”, “pocket depth 4 3 4 bleeding on probing”); a live dictation feedback bar with speech-to-intent preview and a single spoken undo. Dual numbering: FDI 11–48 and Universal 1–32.

FR-2 On-device decision support	Local GGUF inference (Llama-3.2 1B/3B, Gemma-2-2B) bounded by a memory-tier classifier: Tier A ≥6 GB total and ≥2.5 GB free runs Llama-3.2-3B-Instruct Q4_0; Tier B ≥4 GB runs Gemma-2-2B-IT; Tier C below 4 GB runs Llama-3.2-1B. A Clinical Safety Sentinel evaluates recorded allergies, anticoagulant–NSAID bleeding risk and bisphosphonate–extraction MRONJ risk. Optional Gemini multimodal synthesis for radiographs, after local de-identification.
FR-3 3D odontogram	An OpenGL ES / Filament surface-mesh renderer over full maxillary and mandibular arches, with multi-surface selection (mesial, occlusal, distal, buccal, lingual) and a fixed clinical colour language.
FR-4 Workflow & immutability	A Today Queue day-grid with check-in state and encounter timers; a treatment-plan synthesiser phased into emergency, disease control, restorative and maintenance; permanent record locking on signature, amended only by signed dated addenda; and chart PDF export carrying verifiable checksums.
FR-5 Cryptography & audit	SQLCipher AES-256 CBC with keys held in the Android hardware-backed keystore; BiometricPrompt Class 3 lock with a 15-minute idle and 8-hour absolute ceiling; and an append-only trail where each row stores SHA-256 of the previous hash concatenated with timestamp, actor, action and resource, with triggers forbidding update and delete.

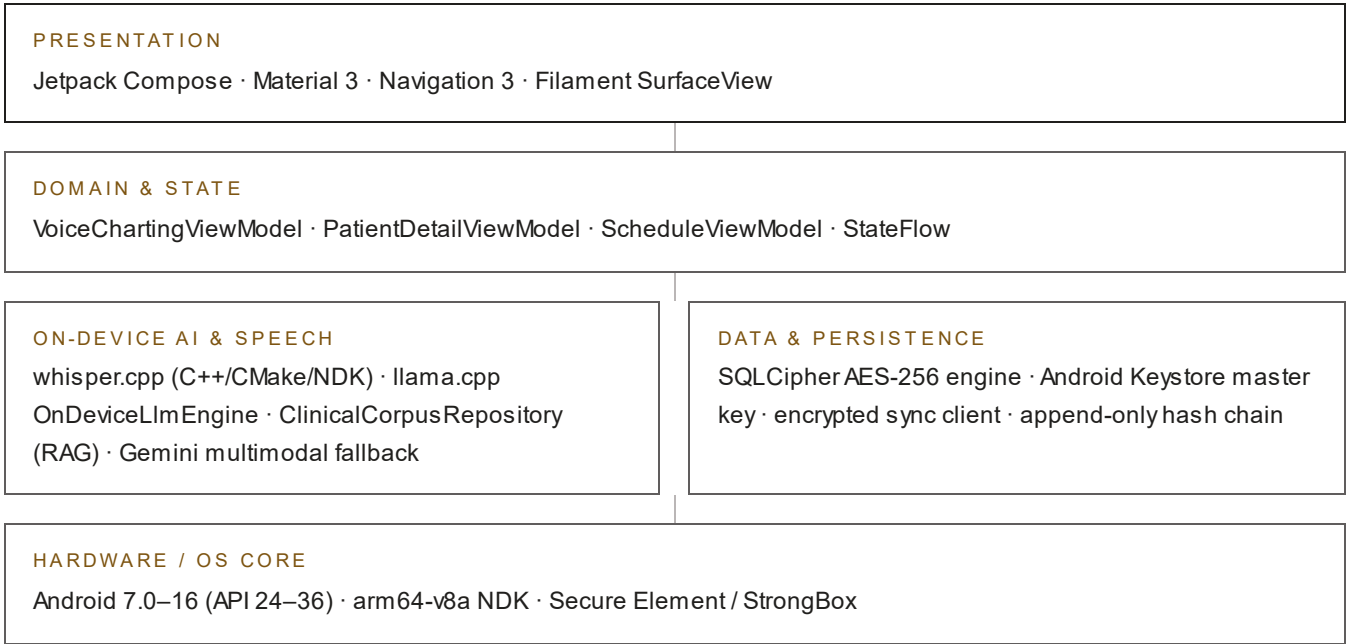
4 · CLINICAL COLOUR LANGUAGE

■ Active caries	■ Sound restoration	■ Defective margin	■ Endodontic obturation
■ Extracted / missing			

5 · NON-FUNCTIONAL BUDGETS

CATEGORY	REQUIREMENT	BUDGET
Speech latency	End of utterance to UI state mutation, quantised Whisper on arm64-v8a	≤ 450 ms
LLM response	Time to first token, and sustained output throughput	≤ 800 ms ≥ 14 tok/s
3D viewport	Filament viewport during rotation, zoom and surface tagging	60 FPS
Security	Unencrypted plaintext written to disk; caches flushed on session end	Zero
Cold launch	Process start to Today Queue on a mid-range device	≤ 1.8 s
Memory ceiling	Peak heap, excluding native mmap of GGUF weights	≤ 480 MB

6 · LAYERED ARCHITECTURE



7 · STACK JUSTIFICATION

SUBSYSTEM	SELECTED TECHNOLOGY	JUSTIFICATION AND CLINICAL TRADE-OFF
UI framework	Jetpack Compose M3	Declarative multi-pane tablet layouts, strict unidirectional data flow, no view-hierarchy inflation lag.
3D engine	Google Filament	Mobile-first physically based rendering; far smaller memory and startup cost than a game engine while still rendering credible enamel.
Speech	whisper.cpp	NEON/AVX SIMD Whisper on CPU with no network dependence — chairside audio never leaves the device.
On-device LLM	llama.cpp, GGUF Q4_0	1B–3B quantised inference inside app memory, giving local assistance without transmitting raw PHI.
Local database	SQLCipher for Android	256-bit AES at rest, transparently meeting HIPAA § 164.312(a)(2)(iv).
Companion portal	Next.js · Neon Postgres	Serverless relational store with pooled connections and edge middleware auth for administrative management.

8 · DATA MODEL

Nine relations carry the clinical record. *Users* manage patients, conduct appointments and author audit rows. *Patients* own tooth conditions, treatment plans, prescriptions and clinical reports; plans own their line items. Two invariants are enforced in the schema rather than the application: a locked plan cannot be mutated, and an audit row can never be updated or deleted.

users	patients	tooth_conditions
id · email · full_name · role · password_hash · salt · is_disabled	id · chart_number · name · date_of_birth · allergies_json · medical_alerts_json	patient_id · tooth_number · numbering_system · surfaces_json · condition_code · severity
treatment_plans	prescriptions	audit_logs
patient_id · clinician_id · status · is_locked · total_cost_cents · locked_at	drug_name · dosage · frequency · duration_days · safety_override_reason	prev_hash · current_hash · actor_id · action_type · resource_type · timestamp

```

CREATE TRIGGER IF NOT EXISTS trg_prevent_locked_plan_mutation
BEFORE UPDATE ON treatment_plans FOR EACH ROW WHEN OLD.is_locked = 1
BEGIN
    SELECT RAISE(ABORT, 'ILLEGAL_MUTATION: Locked treatment plans cannot be
        modified. Create a dated addendum.');
```

```

END;

CREATE TRIGGER IF NOT EXISTS trg_no_delete_audit
BEFORE DELETE ON audit_logs
BEGIN
    SELECT RAISE(FAIL, 'SECURITY_VIOLATION: Audit logs are immutable.');
```

```

END;
```

Figure 1 — Schema-level immutability, enforced by SQLCipher triggers rather than by application code.

```

GET /api/v1/clinical/cds/safety-query      → response
{
  "safety_flag": "HIGH_RISK_WARNING",
  "category": "MRONJ_CONTRAINDICATION",
  "clinical_rationale": "Oral bisphosphonate therapy lasting >3 years
    increases risk of osteonecrosis of the jaw following invasive
    dentoalveolar surgery.",
  "suggested_actions": [
    "Evaluate conservative endodontic therapy if feasible.",
    "Obtain informed consent detailing 0.1–0.21% MRONJ incidence.",
    "Recommend pre- and post-operative chlorhexidine 0.12% rinses."
  ]
}
```

Figure 2 — Cloud fallback decision support. Mutations push over mutual TLS; every payload carries its audit hash.

II Legal & Compliance

PRE-RELEASE · 1.0.0-LEGAL

Practix operates on an offline-first, local-sovereignty architecture. Charting, dictation, demographics and odontogram records are held on the device in a 256-bit AES encrypted SQLCipher database. Protected health information is not monetised, harvested, sold or profiled, and does not reach Practix servers unless a practice explicitly configures cloud synchronisation.

1 · WHAT IS COLLECTED, AND WHERE IT RESTS

CATEGORY OF DATA	METHOD OF PROCESSING	STORAGE LOCATION AND RETENTION
Patient clinical records (PHI / SPDI)	Entered by hand or dictation, processed wholly on device	Local encrypted SQLCipher database; never transmitted unless practice-configured sync is enabled.
Audio dictation streams	Microphone capture into the native Whisper engine	Ephemeral memory buffer, purged on token generation. No raw audio is ever written to disk.
Clinician credentials	BiometricPrompt and script-hashed passwords	Biometric templates stay isolated in the device TEE/StrongBox and are non-exportable.
Diagnostics and crash logs	Opt-in, anonymised system error logs	Ephemeral; stripped of every clinical identifier and chart fragment before transmission.

2 · REGIONAL POSTURE

UNITED STATES

HIPAA & HITECH

A standard Business Associate Agreement governs ePHI under 45 CFR § 164.502(e). Technical safeguards: AES-256 at rest, 15-minute automatic logoff, and hash-chained audit controls over every read, edit and export.

INDIA

DPDP Act 2023

The licensed practice is the Data Fiduciary; Practix is strictly the processor. Processing rests on Section 7 grounds — diagnosis, treatment planning and medical safety — with principal rights served through the clinician console.

EU & UNITED KINGDOM

GDPR / UK DPA

Lawful basis is Article 9(2)(h): preventive medicine and the provision of health treatment. Data remains on the clinician's terminal; contracted backup sits in ISO 27001 / SOC 2 centres inside the EU or UK.

IMPORTANT CLINICAL NOTICE

Decision support, red-flag warnings and on-device model suggestions do not constitute medical diagnoses, definitive treatment directives or clinical prognoses. The algorithms evaluate entered parameters against standardised rules and corpora; they may not account for rare presentations, atypical comorbidities or erroneous input. The clinician must independently cross-examine every suggested interaction, regimen and intervention against current professional guidance — ADA, CDC, BNF or CDSCO.

3 · LICENCE AND RESPONSIBILITY

Licence grant. A non-exclusive, non-transferable, revocable commercial licence to install and run the application on authorised devices, solely for lawful dental practice management, recordkeeping and diagnostic assistance.

Certification. The user warrants they are a licensed dentist, specialist, registered hygienist, or an authorised agent under direct supervision. Practix is not intended for consumer or uncertified self-diagnosis.

Independent judgement. The attending clinician bears sole and exclusive responsibility for verifying chart entries, notations, measurements, prescriptions

and dosages before care is administered or records released. Practix does not practise dentistry.

Prohibited conduct. No reverse engineering of native binaries or proprietary model weights; no bypassing encryption, audit triggers or the biometric overlay; and no recording of patients without the statutory consent required in the jurisdiction.

Liability. The software is provided as is, without warranties of any kind. Aggregate liability is capped at the licence fees actually paid in the twelve months preceding a claim, and indirect, incidental and consequential damages are excluded to the maximum extent the law permits.

4 · STORE IDENTITY & PRIVACY DECLARATION

FIELD	VALUE
App title (Google Play)	Practix: Dental Charting & CDS
Subtitle (App Store)	Voice Odontogram & Clinic EHR
Short description	Offline voice dental charting, 3D odontogram, and on-device clinical decision support.
Categories	Medical (primary) · Productivity / Business (secondary)
Content rating	PEGI 3 / Everyone — medical information reference

DATA TYPE	COLLECTED	DECLARATION
Health & medical data	Yes — local only	Stored solely on the device, not linked to identity across apps, never tracked.
Audio recordings	No	Ephemeral inference in volatile RAM; no audio file is retained.
Identifiers & biometrics	No	Handled exclusively by the device TEE through BiometricPrompt; no biometric data is received.
Analytics & diagnostics	Optional	Anonymised logs stripped of all clinical PHI.

III Operations, QA & Delivery

1.0.0-OPS

PHASE 1 · SPRINTS 1-6 · WEEKS 1-12

Core clinical MVP

Objective: offline operator hardening. 234 story points of aggregate velocity.

PHASE 2 · SPRINTS 7-12 · WEEKS 13-24

Multimodal & cloud

Objective: synchronisation mesh, vision copilot, multi-chair practice.

I · PHASE ONE SPRINT SCHEDULE

SPRINT	EPIC	DELIVERABLES	POINTS
01	Hardware crypto & data core	SQLCipher initialisation from keystore keys; immutable DDL, triggers and hash-chained audit; biometric auto-lock lifecycle observer.	38
02	Speech & JNI pipeline	Native whisper.cpp build via CMake for arm64 and x86_64; circular-buffer audio streaming; deterministic command parser for both numbering systems.	44
03	3D odontogram	Filament integration; mesial-occlusal-distal surface mapping; colour-coded clinical status shaders.	40
04	Decision support core	llama.cpp GGUF runtime; memory allocator and device-tier selector; red-flag checker for allergies, NSAIDs and MRONJ.	46
05	Practice workflows	Today Queue and day-grid schedule; demographics and immutable plans; prescription issuer with override modal.	34
06	Hardening & pilot	Chart PDF generator with cryptographic checksums; ambient operator noise stress testing; production signing and verification.	32
Total	Twelve weeks · six sprints		234

2 · VERIFICATION MATRIX

TEST	CONDITION	EXECUTION	ACCEPTANCE
TC-SP01	Speech under ambient noise	Inject 65–75 dB drill and suction audio while dictating “Tooth 36 MOD composite”.	Word error rate ≤ 4.2%; intent parsing accuracy 100%.
TC-SP02	Rapid voice reversion	Dictate “Tooth 14 caries”, then immediately “undo that”; observe UI and mesh.	State restored in < 200 ms; amber surface flag cleared.
TC-AI01	Dynamic RAM safety	Synthesise a treatment plan on a 4 GB device under background memory pressure.	Gemma-2B or Llama-1B selected; zero OOM termination.
TC-AI02	Contraindication sentinel	Prescribe ketorolac 10 mg to a patient recorded on apixaban.	Modal blocks issuance; written override required.
TC-3D01	Viewport performance	Render 32 tooth meshes; rapid multi-touch rotation, zoom and 50 tag toggles.	≥ 58 FPS sustained; no memory leak after 50 toggles.
TC-SEC1	Storage validation	Pull the sandbox database over ADB and inspect in a hex editor.	SQLite CLI cannot open it; entropy confirms AES.
TC-SEC2	Audit chain integrity	Attempt raw update and delete on the audit table; restart and revalidate.	Triggers raise failure; chain validates 100%.
TC-SEC3	Idle timeout	Leave the application untouched for fifteen minutes and one second.	Overlay appears instantly; keystore key required to resume.

3 · KEY LIFECYCLE & THE HASH CHAIN

O1

Generation

Master keys are generated in the keystore with encrypt and decrypt purposes, GCM block mode and no padding; background database service keys are separated from biometric-bound session keys.

O2

Derivation

The SQLCipher passphrase is derived in unswappable memory and handed to the database as a character array that is zeroed immediately after open.

O3

Revocation

On administrative wipe or ten consecutive failed authentications the keystore entries are deleted, rendering the database cryptographically unrecoverable.

$$Hash_n = HMAC-SHA256 (Hash_{n-1} \parallel timestamp \parallel actor \parallel action \parallel resource)$$

Genesis: $Hash_0 = SHA-256("PRACTIX_CLINICAL_GENESIS_SEED")$. On startup the chain is revalidated row by row; a single failure locks the application into read-only quarantine and alerts the practice administrator.

Three further leakage controls sit alongside: audio buffers are overwritten with zeros the moment a Whisper frame is ingested; clinical endpoints and PDF previews send no-store, no-cache, must-revalidate; and the window carries FLAG_SECURE, keeping clinical screens out of the recents switcher and OS screen capture.

4 · LOCALISATION & ACCESSIBILITY

MARKET	LANGUAGE	NOTATION STANDARD	UNITS & DATE
United States	English (en-US)	Universal numbering, 1–32 / A–T	Metric mg/mL, MM/DD/YYYY
India	English (en-IN), Hindi	FDI World Dental Federation, 11–48	Metric mg/mL, DD/MM/YYYY
United Kingdom	English (en-GB)	Palmer notation / FDI two-digit	Metric mg/mL, DD/MM/YYYY
European Union	German, French	FDI World Dental Federation, 11–48	Metric mg/mL, DD.MM.YYYY

WCAG 2.1 AA CRITERION	LEVEL	IMPLEMENTATION
1.4.3 Contrast (minimum)	Supports	Text on the cream canvas and dark surfaces exceeds 4.5:1, and 7:1 for primary clinical alerts.
2.1.1 Keyboard / switch	Supports	Every tooth, action sheet and modal form is reachable by D-pad and hardware switch with visible focus.
2.5.5 Target size	Supports	Primary targets hold a 48 × 48 dp bounding box for gloved chairside use.
3.3.1 Error identification	Supports	Contraindications and missing fields carry both a high-contrast badge and semantic TalkBack audio.
4.1.2 Name, role, value	Supports	3D surface meshes expose Compose semantics declaring tooth number, surface and clinical state.

IV Business & Intellectual Property

1.0.0-BIZ-IP

STAGE ONE	STAGE TWO	STAGE THREE
Product-led adoption Free student and solo tier; utility earned at the chair.	DSO conversion Multi-chair subscriptions, practice analytics, cloud sync.	Revenue expansion Multimodal add-ons and clearinghouse integrations.

I · IDEAL CUSTOMER PROFILE

SEGMENT	PROFILE	VALUE DRIVER
Solo private dentist	1–2 operatories, general dentistry, \$400k–\$800k collections	Removes chairside charting delay with no IT or server overhead; runs entirely on an Android tablet.
Group practices & DSOs	5–50+ operatories, multi-specialty, \$2M–\$25M collections	Standardised documentation compliance, reduced cross-contamination risk, tamper-evident audit logging.
Academic institutions	Universities, hygiene schools, residency programmes	3D odontogram teaching, structured decision support, dual-notation training.

2 · SUBSCRIPTION TIERS

CAPABILITY	FREE — SOLO / STUDENT	PRO PRACTICE — \$89 / CHAIR / MO	DSO ENTERPRISE — \$149 / CHAIR / MO
Offline voice engine	Standard Whisper	High-precision Whisper	Custom operatory tuning
3D odontogram	Standard quadrant view	Full anatomical mesh with colour codes	Custom mesh and practice branding
Local CDS	Basic allergy checks	Full multi-agent Llama / Gemma engine	Custom DSO clinical rule engine
Active patients	Up to 100	Unlimited	Unlimited
Sync & backup	Manual local export	Multi-device real-time sync	End-to-end encrypted private cloud or on-premise
Audit & security	Standard SQLite	SQLCipher with biometric lockout	Full SIEM and SOC 2 audit export
Support	Community, self-serve	24/5 email and in-app	24/7 dedicated representative with BAA

3 · UNIT ECONOMICS

ACQUISITION COST	LIFETIME VALUE	LTV : CAC	GROSS MARGIN
\$340 → \$195 by year three	\$3,200 → \$5,800 by year three	9.4 : 1 → 29.7 : 1	88% → 92%, net churn under 0.3%

Margin is structurally high because inference happens on the clinician's own hardware: there is no per-utterance speech bill and no per-token reasoning bill. Year-three figures assume referral loops inside dental service organisations and serverless optimisation of the synchronisation path.

4 · CONFIDENTIALITY & INVENTIONS ASSIGNMENT

Summary of the standard contractor instrument. This is a template for engineers and clinical advisors; qualified counsel should review it before execution.

Confidential information. Source, algorithms and build scripts for the native bridges; quantised weights, prompts and fine-tuning corpora; the rendering pipeline, shaders and mesh assets; and internal metrics, strategy, clinician rosters and pricing. Public knowledge, prior lawful possession and independent

development are excluded.

Assignment. Work product is treated as work made for hire; where it is not, all worldwide right, title and interest are assigned irrevocably — patents and applications, copyright and moral rights, trade secrets and know-how, trademarks and trade dress — with a power of attorney to execute filings if a signature cannot be secured.

Models and datasets. GGUF configurations, LoRA weights, checkpoints and prompt assets are exclusively the company's. No external or competing

model may be trained on the company's clinical corpora, dictation recordings or dental datasets.

Covenants. Twelve months of non-solicitation of employees, contractors and clinical advisors; twelve months of non-competition limited to a standalone on-device voice dental charting application.

Return and purge. On termination: hardware and keys returned; all copies of source, weights, local databases and documentation cryptographically overwritten; written certification of compliance within five business days. Governed by Delaware law.

OWNER & DEVELOPER

Devanapally Charan Tej

Sole proprietor and principal developer, Practix

Date _____

FOUR SUITES · ONE RECORD

PRACTIX · CLINICAL WORKSPACE

COMPILED 22 SEPTEMBER 2026